

The Effects of Feedback on Perseverative Errors in Preschool Aged Children

Natalie L. Bohlmann and Larry Fenson
San Diego State University

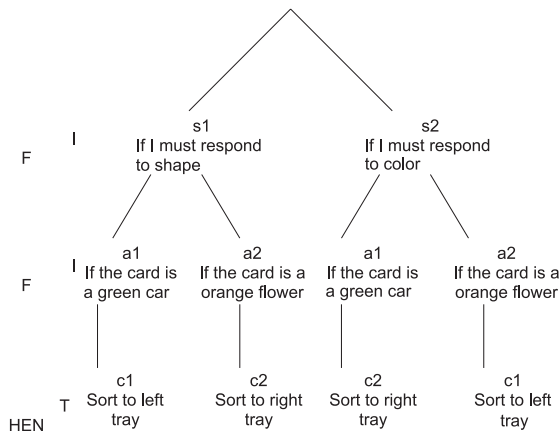
Research using the Dimensional Change Card Sort (DCCS) showed that young children are usually able to sort accurately by an initial rule but are unable to switch to a new rule when the two rules conflict. In 2 experiments, the DCCS was modified to study the effects of feedback on 3- to 5-year-old children in a problem-solving task. In Experiment 1, half of the children in each of two age groups (36 to 44 months and 52 to 60 months) were administered the DCCS task using the standard (no feedback) procedure and the other half received feedback on their post-switch responses. Children who received feedback were able to categorize according to the new (correct) rule, whereas the children in the younger age group who did not receive feedback continued to perseverate. Experiment 2 with 3-year-olds replicated the results from Experiment 1 but found that children's successful performance with feedback on the card-sorting task did not lead to improved performance on the post-switch phase of a subsequent DCCS task. Successful performance under conditions of feedback in both studies implies that 3-year-olds are capable of shifting their response mode from one rule to an alternate rule under conditions that offer clear guidance. Poor performance on the standard version is interpreted to be a reflection of the inability to monitor their own task performance in the absence of clear contextual cues.

One of the tasks that has been used effectively to study features of problem solving in preschool aged children is the Dimensional Change Card Sort (DCCS; Frye, Zelazo, & Palfai, 1995). The challenging aspect of the DCCS is that it places two different pairs of rules in conflict with one another. Children are shown a set of cards with two dimensions such as color and shape (blue and red boats and bears) and are given rules for sorting on one dimension (e.g., color). After a number of tri-

als, they are asked to sort on a different dimension—for example, shape (Frye et al., 1995; Zelazo, Frye, & Rapus, 1996).

When asked to complete a DCCS task, children as young as 3 years of age are able to use the first rules (sort by color) to sort the cards successfully; however, when they are asked to switch to new rules (sort by shape), they perseverate to the first rules learned. Perseveration to the first rules occurs regardless of the order in which the rules are presented or the dimensions used (Frye et al., 1995). By the time a child reaches the age of 5, they are able to switch between the two pairs of rules without any difficulty (Frye et al., 1995; Zelazo et al., 1996).

One interpretation for this perseveration is that 3-year-olds are inflexible in their thinking (Jacques, Zelazo, Kirkham, & Semcesen, 1999). According to Cognitive Complexity and Control (CCC) theory, there are age related changes in the complexity of rule systems that children can represent (Zelazo & Frye, 1997, 1998). Success on the DCCS is predicated on the ability to effectively negotiate a rule hierarchy, shifting from one branch to the other (see Figure 1). As outlined by CCC theory, this apparent “representational inflexibility” of 3-year-olds is attributed to an inability to access higher order rules (s1 and s2) that govern the use of the equivalent and conflicting lower order rules (a1 and a2; Jacques et al., 1999; Zelazo & Frye, 1997, 1998; Zelazo & Jacques, 1996). In this view, 3-year-olds are limited in their representation of rules to *if-then* statements in making judgments correctly (“if red, ... here; if blue, ... there”). To be successful on the DCCS, the



Note: s1 and s2 = setting conditions; a1 and a2 = antecedent conditions; c1 and c2 = consequences

FIGURE 1 The hierarchical structure of rules used on the DCCS as proposed by the CCC theory (adapted from Frye et al., 1995).

child must be able to use a higher order, *if-if-then* rule to make the correct judgment (“if responding to shape, ... if red boat, then sort to left tray; but if responding to color, ... if red boat, then sort to right tray”; Zelazo & Frye, 1998). According to the theory, 3-year-olds know both rules (sort by shape, sort by color) but when these rules are in conflict they are either unable to create or unable to access the setting condition (s1 and s2) in a rule hierarchy to determine which rules to use. By age 5 years, children have apparently developed the ability to adjust their strategy (on the DCCS task) when presented with a new rule.

The strong version of the CCC theory depicts cognitive development as stage-like, inferring that 3-year-olds are incapable of shifting between incompatible pairs of rules (Zelazo & Frye, 1997). However, the developers of the theory have softened their approach, acknowledging that under some conditions children will exhibit more flexibility, and it would be an oversimplification to conclude that 3-year-olds are always incapable of switching (Jacques et al., 1999).

In the standard DCCS task, the rules for sorting are explicit; however, children are given no feedback on their post-switch performance. This raises the possibility that their inability to adopt new rules may result from their failure to monitor their performance rather than from their inability to shift to new rules. On other tasks, researchers have noted that whether young children will detect and attempt to correct their own errors is dependent on whether the task contains sufficiently explicit feedback that an error has been made (DeLoache, Sugarman, & Brown, 1985). In the absence of feedback, there is no compelling reason for them to do so; they may not even be aware that their strategy is incorrect. However, if it is made clear that their actions are not correct, they may recognize that the prior rules are no longer applicable and switch to the new ones. By age 4 to 5 years, children will have acquired a number of metacognitive skills and will have made strides in monitoring their own performance (Gopnik & Wellman, 1994). Yet, 3-year-olds are just beginning this cognitive revolution and may need explicit cues to focus their attention on their performance and on the prevailing rules. Thus, failure to monitor their performance rather than an inability to represent both rules may be what prevents 3-year-olds from shifting to the new rules.

The goal of this research is to explore the effects of feedback on children’s perseverative errors in a rule use task. The DCCS paradigm was used because it clearly taps rule use and has proven to be a workable task with preschool children between the ages of 3 to 5 (Frye et al., 1995; Jacques et al., 1999; Zelazo et al., 1996). In two experiments, the performance of children who received feedback was compared with those who received no such feedback.

Experiment 1 compared the performance of 3 and 5-year-old children administered a standard version of the DCCS task with that of children given feedback on post-switch trials. Experiment 2 assessed whether 3-year-old children who receive feedback on their sorting behaviors learn from the experience and thus perform better on a subsequent standard DCCS task.

EXPERIMENT 1

In the standard DCCS the target and test cards are each sortable on either of two dimensions. Correctly sorting along one dimension requires the child to ignore the other incorrect dimension visible on the target cards. This visual conflict between target and test cards is part of what makes the DCCS difficult for 3-year-old children (Perner & Lang, 2002). When sorting on the first dimension in preswitch sorts, young children are able to ignore the conflicting dimension as irrelevant to the task at hand. Their ability to ignore the irrelevant dimension may be due in part to the experimental procedure in which the experimenter demonstrates correct sorting of the first two preswitch trials. It is when they are asked to switch to the other dimension that 3-year-old children have difficulty. Now, what was irrelevant becomes relevant. Without the demonstration of correct sorting by the experimenter, children have difficulty with the conflicting visual cues. Towse, Redbond, Houston-Price, & Cook (2000) saw improved performance by 3-year-olds when demonstration of post-switch sorting was included along with the verbal explanation of the post-switch rules. Without such information, the child simply sticks to what is known and has worked in the past. If it is made clear to the child which action is correct through feedback on their incorrect sorting behaviors, this may once again allow them to ignore an irrelevant dimension and use the new rules effectively. It was predicted that children who received feedback on the post-switch trials of the DCCS would be able to switch to the new rules.

Method

Participants

A total of 46 children, 23 boys and 23 girls in the age range 36 to 60 months, were tested. Six children (1 boy, 5 girls) were excluded because they failed to cooperate with instructions or because their parent interfered during the task. The final sample of 40 children (22 boys, 18 girls) was divided into a younger group of 20 children (M age = 39.3 months, range = 36–44 months) and an older group of 20 children (M age = 56.3 months, range = 52–60 months). The participants, drawn from a private toddler and preschool program, a local church group, and a university subject pool, were predominantly White middle-class children.

Design

Children were randomly assigned to a feedback group or a no feedback group and received a qualifying phase (assessing the use of a single pair of rules for distinguishing birds and boats), a test phase, and a final sort. Children who did not pass the qualifying phase skipped to the final sort in which there were no defined rules for sorting so that all children (and parents) ended the session with a positive

experience. The final sort was not analyzed and therefore the further details are not included in the experimental procedures.

Apparatus

Three sets of cards (9 cm × 9 cm) and two sorting trays were used. The qualifying phase included 2 target cards (one depicting a bird and the other depicting a boat), which acted as a point of reference for sorting, and 12 test cards, half depicting different birds and half depicting different boats. All subjects used the same set of cards in the qualifying phase. In line with previous DCCS tests, two sets of cards were used in the test phase. Half of the children sorted one set of cards (orange and green cars and flowers) and half sorted another (yellow and blue triangles and circles). For each set, there were 2 target cards (e.g., orange car and green flower or yellow triangle and blue circle) to be matched and 16 test cards (e.g., 8 green cars and 8 orange flowers or 8 yellow circles and 8 blue triangles). All test cards were designed such that cards are sorted differently if one were sorting by shape or by color. All cards were sorted into plain white plastic trays placed adjacent to each other on the table and labeled with the corresponding target cards for each phase of the experimental session.

Procedure

An experimenter met with the children individually in one test session that took approximately 20 min to complete.

Qualifying Phase. Children first participated in a qualifying sort, dividing a dichotomous set into separate trays. Children were presented with a simple pair of rules, all birds go in this tray and all boats go in that tray, and asked to sort task cards according to those rules. The experimenter sorted the first 2 cards (a boat and a bird) into the correct trays, explaining the basis of the sorting to the children. On completion, the experimenter invited the children to sort the remainder of the cards for a total of 10 cards (5 birds and 5 boats) into the two trays labeled with the target cards.

All children received the same pseudorandom order of card presentation restricted such that no more than two cards of the same category were presented consecutively. This card sort served as a criterial measure designed to eliminate children who were not able use a simple rule for sorting or who simply were not willing or able to follow directions. Once children passed the qualifying sort (sorting at least 8 of 10 trials correct), they proceeded to the training phase either in the feedback (experimental) group or in the no feedback (control) group.

Training Phase. The design of the DCCS task for the control group duplicated Zelazo and colleagues' experimental condition (Frye et al., 1995; Zelazo et al., 1996) with a pre- and postswitch phase.

Preswitch trials; all children. In the preswitch phase, children were shown two target cards (e.g., an orange car and a green flower). The experimenter labeled these target cards in terms of both their color and their shape (e.g., "Here's an orange car"). Children were told the rules for sorting and asked to sort cards into two trays according to a single dimension (color):

Now we are going to play a different game. This is the color game. The color game is different. In the color game we put all the pictures that are the same color together. In the color game, all the orange ones go in this box, and all the green ones go in that box. Only orange go in this box and only green go in that box. If it is an orange one it goes here but if it is a green one it goes there. I will do the first two cards to show you how to play the game and then it is your turn.

The experimenter sorted two cards (a green car and an orange flower), explaining the basis for the behavior to the children. The experimenter then invited the children to sort the cards: "Now I am going to show you some orange ones and some green ones, and it is your turn to play the game."

The experimenter showed the children six cards (green cars and orange flowers) drawn from a pseudorandom sequence with the constraint that no more than two cards of the same type were presented in a row. For each card, the experimenter restated the rules ("if it is an orange one then it goes here, if it is a green one then it goes there"), labeled the card by both shape and color ("this is a green car"), and asked the children where it should be placed ("where does it go?"). The experimenter then gave the card to the children to place face down in the tray. Following each trial, the experimenter told them if they were correct or incorrect; "That's right" or "No, that's not right." The experimenter then said, "Are you ready for another one?"

The dimension (color or shape) that was relevant during the preswitch phase was counterbalanced and crossed with age and with the set of cards that children received.

Test Phase

Postswitch trials; no feedback (control) group. After six trials, the experimenter stopped the game and instructed the children on playing a new game according to the alternate dimension. They were told the new rules and asked to sort cards according to the alternate dimension (shape):

Okay, that was a fun game but we don't want to play that game any more. Now we are going to play a new game. This is called the shape game. The shape game is different from the color game. In the shape game all the cars go here in this tray and all the flowers go in that tray. We don't put any cars in that box. No way, all the cars go in this box. Remember, only cars go in this tray and only flowers go in that tray. This is the shape game. Ok, I am going to show you some flower ones and some car ones and it is your turn to play the game.

Children were asked to sort on eight postswitch trials that were identical to the preswitch trials except that on each trial the children were told the rules for sorting by the other dimension ("if it is a car then it goes here, if it is a flower then it goes there"). No feedback was given. Following each trial, the experimenter repeated the rules and said, "Are you ready for another one?" This proceeded until all 8 postswitch cards were sorted or until the children answered five of the last six sorts correctly to a maximum of 16 cards sorted.

Postswitch trials; feedback (experimental) group. The feedback group followed the same procedures as the no feedback group except that following each trial, the experimenter told the children if they were correct or incorrect; "Yes, that's right, all the _____ go here. You're good at the (color or shape) game." or "No, that's not right. Remember, in the (color or shape) game all the _____ go here." While giving verbal feedback, the experimenter turned the card face up and moved it to the correct tray. The experimenter then turned the card faced down in the correct box, repeated the rules and asked, "Are you ready for another?" Trials proceeded until all 8 post-switch cards were sorted or until the children answered 5 of the last 6 sorts correctly to a maximum of 16 cards sorted.

Results

All of the children passed the criterial sort (at least 8 of 10 trials correct) and the preswitch phase (responding correctly on at least 5 of 6 trials). Passing the postswitch phase required correctly sorting on 5 of 6 consecutive trials within 16 possible trials. Fifty-five percent of children 36 to 44 months and 85% of children 52 to 60 months passed the postswitch phase.

To assess the effects of feedback in the postswitch phase, up to 16 trials were administered to all participants. A child who met criterion on Trial 5 was able to switch immediately to the new rule without feedback. Trial 6 was the earliest a child who received negative feedback ("No, that's not right.") on trial 1 could meet criterion.

Only 1 of the 10 children 36 to 44 months of age who did not receive feedback on their responses met criterion, whereas all 10 of the children in the same age

group who received feedback did meet criterion ($M = \text{Trial } 6.4$, $SD = 1.27$). The difference between groups for children 52 to 60 months was less pronounced. Seven children 52 to 60 months of age who did not receive feedback (70%) met criterion and three children did not. All 10 of the children in the older group who received feedback met criterion ($M = \text{Trial } 5.3$, $SD = .68$).

A large percentage (60%) of the children who passed or failed the postswitch phase either succeeded or erred on all trials in that phase. Consequently, nonparametric categorical analysis was used to analyze the data. A chi-square test showed that significantly more children in the feedback group met criterion, $\chi^2(1, N = 40) = 17.143$, $p < .05$, $\phi^2 = .429$, and that performance on the postswitch phase was dependent on feedback for the children in the younger age group, $\chi^2(1, N = 20) = 16.36$, $p < .05$, $\phi^2 = .819$, but not for children in the older age group, $\chi^2(1, N = 20) = 3.53$, $p > .05$, $\phi^2 = .176$. The latter result can be attributed to a ceiling effect, in that a high percentage of children in the feedback group at the older age (52 – 60 months) passed the postswitch phase. This pattern of results on the postswitch phase was not influenced by target set (triangle or circles or flowers or cars), $\chi^2(1, N = 40) = .043$, $p > .05$, $\phi^2 = .001$; by dimension (color or shape) in the preswitch phase, $\chi^2(1, N = 40) = .077$, $p > .05$, $\phi^2 = .002$; or by gender, $\chi^2(1, N = 40) = .173$, $p > .05$, $\phi^2 = .004$.

Most children in the younger age group who received feedback were able to switch to the new rules after only one feedback trial. Five children met criterion by Trial 6 and therefore only needed one feedback trial to switch to the new rules. Only 3 children in this age group required more than one trial of feedback and no one needed more than four feedback trials to switch to the new rules.

The pattern of results is similar for children 52 to 60 months of age. Eight of the 10 children assigned to the feedback condition in this age group did not need feedback to switch to the new rules and met criterion by Trial 5. Of the 2 children who benefited from feedback, 1 child switched to the new rules after one trial of feedback and the other child only needed two feedback trials.

Examination of the data under conditions of feedback revealed no difference in performance attributable to age, with 100% of the children in both age groups passing. For the nonfeedback condition, however, there was an age effect, $\chi^2(1, N = 20) = 7.5$, $p < .05$, $\phi^2 = .375$. This significant difference with respect to age for the non-feedback group replicates previous findings by Zelazo and colleagues (Frye et al., 1995; Jacques et al., 1999; Zelazo et al., 1996).

Discussion

Previous studies using the DCCS have demonstrated that 3-year-olds have trouble switching from one pair of rules to another when two pairs of rules are placed in conflict. Under such conditions they tend to perseverate to the first rules learned.

These results show that 3-year-old children can shift rules appropriately if they are given feedback on their incorrect responses.

In the DCCS task, for each rule there is a matching action sequence that correctly expresses the rule. When two pairs of rules are placed in conflict with one another as on the DCCS, to respond effectively, children must monitor their actions, checking to confirm that these actions match the corresponding rule. These results suggest that poor performance on the standard version of the DCCS may be the result of the 3-year-olds' inability to monitor their task performance in the absence of clear contextual information. The breakdown in the task may not reflect an inability to switch to a new rule *per se* but rather an inability to evaluate the outcomes of their actions in comparison to that rule when conflicting information is present; that is, to realize that the deployed rule was not correct. When feedback is provided, 3-year-olds recognize their error and make the shift to the appropriate rule, most children switching after only one feedback trial.

The results can be interpreted in multiple ways. The ability to switch to the alternate dimension on the DCCS can be interpreted as a demonstration of the 3-year-olds' ability to traverse a rule hierarchy (e.g., if *s1*, then *a1*, then *c1*, whereas if *s2*, then *a1*, then *c2*) under the right conditions (Figure 1). Conversely, if indeed a rule hierarchy is created it may be argued that providing feedback as to the incorrectness of their response does not allow them to traverse the hierarchy but rather prompts them to jump to the other rule branch (i.e., from *s1* to *s2*). If feedback simply moves children to the other branch of the tree structure, then it overstates the power of feedback to conclude that passing the feedback condition demonstrates children's "representational flexibility" as defined by Zelazo and his colleagues (Zelazo et al., 1996).

One type of evidence that could more convincingly demonstrate flexibility in 3-year-olds would be the ability to switch to the alternate rule (from *s1* to *s2*) without assistance on a subsequent card sort. Having had the experience of successfully switching between rules may lead to positive transfer for young children on a second card-sorting task. If learning does occur, then we would expect sorting on a second DCCS to improve. In general, few studies have focused on transfer of learning in preschoolers; the second experiment attempted to address this question.

EXPERIMENT 2

The purpose of this study is to examine whether children who receive feedback on post-switch sorts and therefore switch to the new rule learn from this experience. It is predicted that children who receive feedback on postswitch trials of the DCCS will switch without feedback on a subsequent card sort.

Method

Participants

Thirty-two children, 18 boys and 14 girls in the age range 36 to 44 months, were tested. Seven children (5 boys, 2 girls) were excluded because they failed to pass the preswitch phase, failed to cooperate with instructions, or because their parent interfered during the task. One boy in the feedback group also was excluded because he did not pass the postswitch sort on the first DCCS. The final analysis included 24 children (12 boys, 12 girls, *M* age = 39.46 months, range = 36 to 43 months). The participants were drawn from a university affiliated preschool program and a university subject pool and were predominantly White middle-class children.

Design

Children were randomly assigned to a feedback group or a no feedback group, for a training phase (DCCS 1), a test phase (DCCS 2), and a final sort with positive feedback. Both the no feedback and the feedback groups participated in all phases. As in Experiment 1, only children who met criterion on the preswitch phase were tested on the postswitch phase.

Apparatus

Two full sets of cards (9 cm × 9 cm) and two sorting trays were used. Because the pattern of results on the postswitch phase in Experiment 1 was not influenced by target set (triangle or circles or flowers or cars) all participants used the same full set of sorting cards. The design of the cards were the same as that used in Experiment 1 and included orange and green flowers and cars on the first DCCS and yellow and blue hearts and stars on the second DCCS.

Procedure

An experimenter met with the children individually in one test session that took approximately 20 min to complete. The experiment began with the introduction of the first DCCS in which half of the children received the standard version of the DCCS and half of the children received feedback on postswitch trials as outlined in Experiment 1. Following the first DCCS, participants were then invited to play another game with a new set of cards. All children were then given a second standard DCCS without feedback on postswitch sorts.

Results

Criterion for inclusion in the study was passage of the preswitch phase on the first DCCS. Passage was defined as responding correctly on at least 5 of 6 consecutive

trials. Two children were excluded and replaced based on this criterion. Because most children in the first experiment switched to the new rules with only 1 trial of feedback, passing the postswitch phase required correctly sorting on 5 of 6 consecutive trials within 8 possible trials. With this reduction in feedback trials from 16 to 8, one boy in the feedback group failed to pass the postswitch phase.

The final analysis included 24 children, 12 children who received feedback and passed the postswitch phase in the first DCCS and 12 children who did not receive feedback on postswitch trials of the first DCCS.

Only 2 of the 12 children who did not receive feedback on their responses on the first DCCS met criterion on postswitch trials on the second card sort. Similarly, 3 of the 12 children who received feedback on the first card sort met postswitch criterion on the second DCCS.

A large percentage (62.5%) of the children who passed or failed the postswitch phase on the second card-sorting task either succeeded or erred on all trials in that phase. Consequently, nonparametric categorical analysis was used to analyze the data. A chi-square test showed that performance on the postswitch phase of the second card sort was not dependent on feedback in the first card sort for the children, $\chi^2(1, N = 24) = .253, p > .05, \phi^2 = .011$. This pattern of results on the postswitch phase was not influenced by dimension (color or shape) in the preswitch phase, $\chi^2(1, N = 24) = .253, p > .05, \phi^2 = .011$, or by gender, $\chi^2(1, N = 24) = 2.27, p > .05, \phi^2 = .095$.

Discussion

Experiment 2 explored whether children given feedback that enabled them to successfully shift to the new rule in the postswitch phase of the DCCS task would be more likely to shift to the new rule in the postswitch phase of a new DCCS task. The results of Experiment 2 replicate those of Experiment 1 in demonstrating the effectiveness of feedback given to 3-year-olds on successfully switching to the correct dimension on the DCCS task. However, feedback-aided success on the first DCCS did not lead to successful switching on a second DCCS task. That is, there was no carryover effect. Children who were helped by feedback were no more likely to switch in the absence of feedback on a subsequent card sort.

It has been argued that providing feedback on the DCCS changes the task from a *rule use* task to a *rule-learning* task (Zelazo & Jacques, 1996). If this is true, then children who receive feedback on their perseverative errors and therefore switch to the new rule might be expected to learn from the experience. If learning about a higher order rule were to occur, we might expect that children who are given feedback on one DCCS task would perform better on a subsequent DCCS even if feedback were not given. This expectation was not confirmed by the results of Experiment 2. It is possible that more extensive experience in shifting to a different

dimension would produce transfer to a new DCCS task (such as was demonstrated by the extensive training conducted by Kloo & Perner, 2003 in Experiment 2).

Other researchers have demonstrated other contextual cues that influence DCCS success in younger children (e.g., Perner & Lang, 2002; Towse et al., 2000). Problems children experience on the DCCS may arise from the salience of the preswitch features or from the experience of ignoring the irrelevant features. When the relevant attributes are emphasized, children can be coaxed into using the postswitch rule. Providing children with feedback on their incorrect sorting behaviors on postswitch trials may help children to focus on the relevant features of the task and successfully switch to the new rule.

GENERAL DISCUSSION

Two studies of the effects of feedback on the performance of 3-year-olds on the DCCS task show that when children are given feedback on the postswitch trials, performance improves with most perseverative children switching after one or two trials. The results call into question the view that 3-year-olds are inflexible in their thinking. At the same time, switching to the new dimension under conditions of feedback does not conclusively demonstrate that 3-year-old children successfully create and traverse a rule hierarchy on the DCCS task in the way that 5-year-olds do—as shown in prior studies and in Experiment 1. The fact that 3-year-olds can switch with feedback (Experiments 1 and 2) does not mean this additional scaffolding permits them to do the same thing as 5-year-olds. Rather than traverse a rule hierarchy, it may be that feedback simply orients them to the new rule.

Although feedback allows 3-year-olds to successfully switch to a new rule, the fact that feedback does not facilitate accommodation to the new rule in a second DCCS task indicates that the effects of feedback are contextual and transient. It may be that feedback simply makes salient the cues necessary for correctly acting on the relevant rules to use. The transitory nature of the effect suggests that feedback has a cueing effect. In each situation, children are responding to situational cues and not relating one condition to another. Without feedback—or some other form of modeling of the correct actions—(see Towse et al., 2000, Exp. 1) children may be confused as to which dimensions are relevant.

Additional inquiry is needed to evaluate the factors that most effectively aid the performance of very young children in rule-governed tasks like the DCCS. Why, for example, is it more effective to call attention to errors than to explain that “we are going to play a new game”? Perhaps input in the form of behavioral correction may be more salient than simple description of rules in focusing attention on the relevant properties of the task.

ACKNOWLEDGMENTS

Natalie L. Bohlmann is now at Department of Psychology, University of California, Berkeley.

Experiment 1 is based on data collected for Natalie Bohlmann's master's thesis at San Diego State University and was presented in poster form at the Society for Research in Child Development 2001 biennial meeting in Minneapolis, Minnesota. Experiment 2 was conducted at the University of California, Berkeley.

REFERENCES

- DeLoache, J., Sugarman, S., & Brown, A. (1985). The development of error correction strategies in young children's manipulative play. *Child Development*, 56, 928-939.
- Frye, D., Zelazo, P. D., & Palfai, T. (1995). Theory of mind and rule-based reasoning. *Cognitive Development*, 10, 483-527.
- Gopnik, A., & Wellman, H. M. (1994). The theory theory. In L. A. Hirshfeld & S. A. Gelman (Eds.), *Mapping the mind* (pp. 257-293). New York: Cambridge University Press.
- Jacques, S., Zelazo, P. D., Kirkham, N. Z., & Semcesen, T. K. (1999). Rule selection versus rule execution in preschoolers: An error-detection approach. *Developmental Psychology*, 35, 770-780.
- Kloo, D., & Perner, J. (2003). Training transfer between card sorting and false belief understanding: Helping children apply conflicting descriptions. *Child Development*, 74, 1823-1839.
- Perner, J., & Lang, B. (2002). What causes 3-year-olds' difficulty in the dimensional change card sorting task? *Infant and Child Development*, 11, 93-105.
- Towse, J., Redbond, J., Houston-Price, C., & Cook, S. (2000). Understanding the dimensional change card sort. Perspectives from task success and failure. *Cognitive Development*, 15, 347-365.
- Zelazo, P. D., & Frye, D. (1997). Cognitive complexity and control: A theory of the development of deliberate reasoning and intentional action. In M. Stamenov (Ed.), *Language structure, discourse, and the access to consciousness* (pp. 113-153). Amsterdam: John Benjamins.
- Zelazo, P. D., & Frye, D. (1998). Cognitive complexity and control: II. The development of executive function in childhood. *Current Directions in Psychological Science*, 7, 121-126.
- Zelazo, P. D., Frye, D., & Rapus, T. (1996). An age-related dissociation between knowing rules and using them. *Cognitive Development*, 11, 37-63.
- Zelazo, P. D., & Jacques, S. (1996). Children's rule use: Representation, reflection and cognitive control. *Annals of Child Development*, 12, 119-288.

Copyright of Journal of Cognition & Development is the property of Lawrence Erlbaum Associates and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.